jQuery

SYAI&DS-A

Introduction to jQuery

1. jQuery is a lightweight, "write less, do more", JavaScript library.
2. jQuery is an open-source JavaScript library that simplifies the interactions between an HTML/CSS document
3. jQuery is a fast, small, and feature-rich JavaScript library.
4. The purpose of jQuery is to make it much easier to use JavaScript on your website.
5. jQuery is cross-platform.

jQuery Features

1. Event handling: Simplify event handling such as click, hover, submit, etc.
2. Animations: Create smooth animations and effects.
3. jQuery is simple and has powerful features
4. It helps us to manipulate DOM (Document Object Model) elements
5. Provides event methods to trigger and respond to an events on a html page such as mouse click, keypress etc

Advantages of jQuery

- Wide range of plug-ins that allows developers to create plug-ins on top of the JavaScript library.
- Large development community.
- It is a lot easier to use compared to standard javascript and other javascript libraries.
- It lets users develop Ajax templates with ease. Ajax enables a sleeker interface where actions can be performed on pages without requiring the entire page to be reloaded.
- Being Light weight and a powerful chaining capabilities makes it more strong.

Adding jQuery to Your Web Pages

There are several ways to start using jQuery on your web site.

- Use the Google-hosted/ Microsoft-hosted content delivery network (CDN) to include a version of jQuery.
- Download own version of jQuery from jQuery.com and host it on own server or local filesystem.

jQuery Syntax

- The jQuery syntax is tailor-made for selecting HTML elements and performing some action on the element(s).
- $(selector).action()
  - $ sign: It grants access to jQuery.
  - (selector): It is used to find HTML elements.
  - jQuery action(): It is used to perform actions on the elements.

Examples

- $(this).hide() - hides the current element.
- $("p").hide() - hides all <p> elements.
- $(".test").hide() - hides all elements with class="test".
- $("#test").hide() - hides the element with id="test

Document Ready Event:

```
$(document).ready(function(){
  $(selector).action()       // jQuery methods go here...
});
```

1. All jQuery methods should be inside a document ready event to prevent any jQuery code from running before the document is finished loading

2. The factory function $() is a synonym of jQuery() function.

Document Ready Event:

if methods are run before the document is fully loaded then following actions that can fail

1. Trying to hide an element that is not created yet
2. Trying to get the size of an image that is not loaded yet

jQuery Selectors

jQuery selectors allow you to select and manipulate HTML element(s).

1. HTML element Name
2. Element ID
3. Element Class
4. Element attribute name
5. Element attribute value
6. Many more criteria

• jQuery selectors are used to "find" (or select) HTML elements based on their name, id, classes, types, attributes, values of attributes and much more.

• It's based on the existing CSS Selectors, and in addition, it has some own custom selectors.

• All selectors in jQuery start with the dollar sign and parentheses: $()

jQuery Selectors Syntax

```
$(document).ready(function(){
    $(selector)
});
```

The element Selector

The jQuery element selector selects elements based on the element name.

You can select all <p> elements on a page like this:

```
$(document).ready(function(){
    $("Html Element Name")
});
```

```html
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("button").click(function(){
    $("p").hide();
  });
});
</script>
</head>
<body>
<h2>This is a heading</h2>
<p>This is a paragraph.</p>
<p>This is another paragraph.</p>
<button>Click me to hide paragraphs</button>
</body>
</html>
```

The #id Selector

The jQuery #id selector uses the id attribute of an HTML tag to find the specific element.

An id should be unique within a page, so you should use the #id selector when you want to find a single, unique element.

```
$(document).ready(function(){
    $("#id of the element")
});
```

```html
<!doctype html>
<html>
<head>
<title>The jQuery Example</title>
<script src="https://www.tutorialspoint.com/jquery/jquery-3.6.0.js">
</script>
<script>
   $(document).ready(function() {
      $("#foo").css("background-color", "yellow");
   });
</script>
</head>
<body>
   <h1>jQuery #id Selector</h1>

   <p id="foo">This is foo p tag</p>
   <span id="bar">This is bar span tag</span>
   <div id="bill">This is bill div tag</div>
</body>
</html>
```

The #class Selector

The jQuery .class selector selects HTML element(s) based on the element class attribute.

```
$(document).ready(function(){
    $(".class of the element")
});
```

```html
<!doctype html>
<html>
<head>
<title>The jQuery Example</title>
<script src = "https://www.tutorialspoint.com/jquery/jquery-3.6.0.js">
</script>
<script>
   $(document).ready(function() {
      $(".foo").css("background-color", "yellow");
   });
</script>
</head>
<body>
   <h1>jQuery .class Selector</h1>
   <p class="foo">Web Technology</p>
   <p class="foo">jQuery</p>
   <span class="bar">Second Year</span>
   <div class="bill">Batch</div>
</body>
</html>
```

```html
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("button").click(function(){
    $(".test").hide();
  });
});
</script>
</head>
<body>
<h2 class="test">This is a heading</h2>
<p class="test">This is a paragraph.</p>
<p>This is another paragraph.</p>
<button>Click me</button>
</body>
</html>
```

jQuery Event Methods

jQuery events are the actions that can be detected by your web application. They are used to create dynamic web pages. An event shows the exact moment when something happens.

Examples of events.

A mouse click

An HTML form submission

A web page loading

A keystroke on the keyboard

Scrolling of the web page etc.

Mouse Events	Keyboard Events	Form Events	Document Events
click	keypress	submit	load
dblclick	keydown	change	resize
hover	keyup	select	scroll
mousedown		blur	unload
mouseup		focusin	ready

Syntax

```
$("div").click(function(){
   // jQuery code goes here
});
```

```
$("div").bind('click', function(){
   // jQuery code goes here
});
```

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.goleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("p").click(function(){
    $(this).hide();
  });
});
</script>
</head>
<body>
<p>If you click on me I will disappear.</p>
<p>Click me away!</p>
<p>Click me too!</p>
</body>
</html>
```

```html
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("p").dblclick(function(){
    $(this).hide();
  });
});
</script>
</head>
<body>

<p>If you double-click on me, I will disappear.</p>
<p>Click me away!</p>
<p>Click me too!</p>

</body>
</html>
```

How can we change styles when certain events got triggered with the help of JQuery.

- We can apply styles on elements by using either the

1. **JQuery CSS Classes** or the
2. **JQuery .css() method**.

Approach 1: Using JQuery CSS Classes..

- This approaches consists of using methods such as the addClass() method, removeClass() method and the toggleClass() method

- We can create some classes with some css rules and add or remove them using JQuery as per the requirement.

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <script src=
"https://code.jquery.com/jquery-3.6.0.min.js">
    </script>
    <style>
        #poll {
            position: absolute;
            bottom: 15%;
        }

        #submit {
            position: absolute;
            bottom: 10%;
        }

        .safe {
            background-color: rgb(6, 161, 6);
            width: 20%;
            height: 30%;
        }
```

```html
        .danger {
            background-color: rgb(226, 19, 19);
            width: 20%;
            height: 60%;
        }
    </style>
</head>

<body>

    <h3>How can you apply a style on an
        element using jQuery?</h3>
    <div id="poll" class="safe"></div>
    <button id="submit">Alarm</button>
    <script>
        $("#submit").click(function () {
            // We can either remove the old
            // class and add a new one
            // $("#poll").removeClass("safe").addClass("danger");

            // Otherwise, we can toggle the class
            $("#poll").toggleClass("danger");
        })
    </script>
</body>

</html>
```

Approach 2: Using the JQuery css() method:

- In this method, instead of creating separate classes inside the stylesheet, you can directly set styles from javascript itself.
- The **css() method** can be used to either return a single property based on the given property or set a single or multiple CSS properties.

```
<!DOCTYPE html>
<html lang="en">

<head>
    <script src=
"https://code.jquery.com/jquery-3.6.0.min.js">
    </script>
</head>

<body>
    .
    <h3>How can you apply a style on an
        element using jQuery?</h3>
    <b>Telephone Number Verification</b>
    <br><br>
    <input type="number" id="phone"
           placeholder="Enter the Phone No." />
    <span id="warning"></span><br><br>
    <button id="submit">Check</button>

    <script>
        $("#submit").click(function () {

            // Regular expression to check
            // for an indian phone number
            const regexExp = /^[6-9]\d{9}$/gi;
            const phone = $("#phone").val();
```

```
            // Add or remove classes based on
            // the regular expression check
            if (!regexExp.test(phone)) {
                $("#warning").text("It's wrong try again!");
                $("#warning").css("color", "red");
                $("#phone").css("background", "red");
            } else {
                $("#warning").text("It's correct");
                $("#warning").css("color", "green");
                $("#phone").css("background", "green");
            }
        });
    </script>
</body>

</html>
```

How to create an HTML element using jQuery ?

- To create and append the HTML element, we use the jQuery append() method.
- The jQuery append() method is used to insert some content at the end of the selected elements.
- Syntax:

```
$(selector).append( content, function(index, html) )
```

Parameters:

- **content:** It is a required parameter and is used to specify the content which is to be inserted at the end of selected elements. The possible value of contents is HTML elements, jQuery objects, and DOM elements.
- **function(index, html):** It is an optional parameter and is used to specify the function that will return the content to be inserted.
  - **index:** It is used to return the index position of the element.
  - **html:** It is used to return the current HTML of the selected element.

```
<!DOCTYPE html>
<html>
<head>
    <title>
        How to create an HTML element using jQuery?
    </title>
    <script src="
https://ajax.googleapis.com/ajax/libs/jquery/3.3.1/jquery.min.js">
    </script>
    <!-- Script to add div element in the HTML document -->
    <script>
        $(document).ready(function() {

            $("button").click(function() {
                $(".append").append(
'<div class="added">New HTML element added</div>');
            });
        });
    </script>
    <!-- Style to use on div element -->
```

```html
    <style>
        .added {
            padding: 20px;
            margin-top: 20px;
            background: green;
            color: white;
            display: inline-block;
        }
    </style>
</head>
<body>
    <center>
        <h1 style="color: green;">GeeksforGeeks</h1>

        <h3>
            How to create an HTML element using jQuery?
        </h3>

        <button id="append">Append New HTML Element</button>

        <div class="append"></div>
    </center>
</body>
</html>
```

jQuery - Remove Elements

Remove Elements/Content

- To remove elements and content, there are mainly two jQuery methods:

1. remove() - Removes the selected element (and its child elements)
2. empty() - Removes the child elements from the selected element

```html
<html>
<head>
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/
jquery.min.js"></script>
<script>
$(document).ready(function(){
  $("button").click(function(){
    $("#div1").remove();
  });
});
</script>
</head>
<body>

<div id="div1" style="height:100px;width:300px;border:1px solid
black;background-color:yellow;">

This is some text in the div.
<p>This is a paragraph in the div.</p>
<p>This is another paragraph in the div.</p>

</div>
<br>

<button>Remove div element</button>

</body>
```

This is some text in the div.

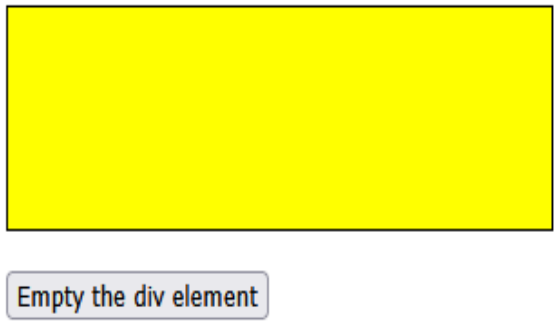
This is a paragraph in the div.

This is another paragraph in the div.

Remove div element

jQuery empty() Method

```html
<html>
<head>
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>
<script>
$(document).ready(function(){
  $("button").click(function(){
    $("#div1").empty();
  });
});
</script>
</head>
<body>

<div id="div1" style="height:100px;width:300px;border:1px solid black;background-color:yellow;">

This is some text in the div.
<p>This is a paragraph in the div.</p>
<p>This is another paragraph in the div.</p>

</div>
<br>

<button>Empty the div element</button>

</body>
</html>
```



Filter the Elements to be Removed

- The jQuery remove() method also accepts parameter, which allows you to filter the elements to be removed.
- The parameter can be any of the jQuery selector syntaxes.

```html
<!DOCTYPE html>
<html>
<head>
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>
<script>
$(document).ready(function(){
  $("button").click(function(){
    $("p").remove(".test, .demo");
  });
});
</script>
<style>
.test {
  color: red;
  font-size: 20px;
}

.demo {
  color: green;
  font-size: 25px;
}
</style>
</head>
<body>
```

This is a paragraph.

This is p element with class="test".

This is p element with class="test".

This is p element with class="demo".

Remove all p elements with class="test" and class="demo"

```html
<p>This is a paragraph.</p>
<p class="test">This is p element with class="test".</p>
<p class="test">This is p element with class="test".</p>
<p class="demo">This is p element with class="demo".</p>

<button>Remove all p elements with class="test" and class="demo"</button>

</body>
</html>
```